

the operators can transmit only seven SD channels. But by using DVB-S2 modulation combined with MPEG-4 (H.264) encoding, operators will be able to transmit 26 SD channels to the same transponder.

While standardisation of software application processor interface (API) plays a key role in easing the software integration process to multiple complex modules, the underlying semiconductors need to work as a 'megawatt engine' to drive software complexities.

Semiconductor technologies help in making available high-quality TV services to the customers at a very affordable cost. In order to make HD MPEG-4 services affordable, continued advances in design and smaller process geometry (nm) are the need of the hour. To comply with this trend semiconductor players are integrating more functions like demodulation, transcoding, high-end (3D) graphics along with MPEG decoders, all rolled into a single chip. Companies like NXP have also chosen to move to 45nm process geometry as smaller process geometry has advantages like lower power, lower costs and higher integration possibilities.

High level of integration inside the chipset allows lesser time to market as the integration time and hassle have both been reduced. It is expected that in the coming years semiconductor technology would advance considerably to meet the growing demand in the digital TV domain.

STB growth in India

A couple of years back, STBs were associated with CAS under a tough regulatory regime imposed by the government. Doordarshan had started a free-to-air service (DD Direct) which made TV penetration into remote villages possible. Today STB is increasingly seen as a technology easily deployable in living rooms, fuelling consumer appetite for high-quality digital pictures.

Fast growth in STB sales in India

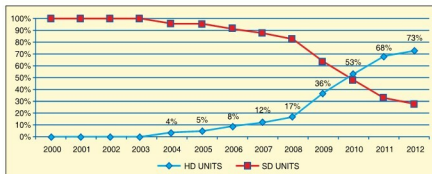


Fig. 1: High definition vs standard definition worldwide

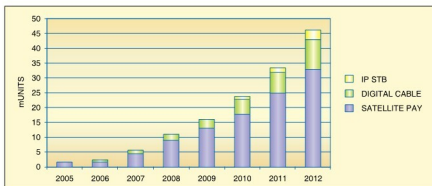


Fig. 2: STB projections in the next three years

is also being driven by increased sale of LCD TVs in India. A crossover in 2012 between the ramp-down in CRT demand and a ramp-up of LCD TV shipments have been predicted.

Other growth drivers for STBs include:

1. Availability of more channels
2. Availability of online movie channels
3. TV services in remote areas via DTH
4. Affordability – technically and commercially
5. Digital picture quality
6. Emerging acceptance of online shopping and gaming

In the next three to four years, CAGR is expected to be about 40 per cent, giving huge opportunities to the STB industry. Fig. 2 gives the forecast up to 2012, based on various reports and some extrapolation.

While more and more operators have joined the fray, the size of the market has only increased due to competitive packages and addition

of attractive programmes/channels. DTH subscription crossed the 1 million-mark in October 2008 for the first time ever.

Local vs imports

Operators are shelling out at least Rs 1000 per connection in the form of subsidies, hoping to recover costs through the economies of scale which would mean a high subscription base. The current strategy of most operators is to acquire the broadest customer base and later make money through value-added services like video-on-demand, target advertisements, HD services, etc.

No doubt manufacturing of STBs in India is lagging behind. For Doordarshan free-to-air STBs are manufactured in India by contract manufacturers, but they have very little experience in making CAS boxes. Jabil Circuits, Chennai, has been manufacturing STBs for Tata Sky.

Unfortunately, more than 95 per cent of the STBs are imported. Of these,